

## GENT INDUSTRIE-ZONE, Belgium

WMO: 064310

Lat: 51.1806N

Lon: 3.8042E

Elev: 26

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-18

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 22.8       | 26.4 | 16.3                          | 12.6 | 26.0 | 20.6 | 15.4 | 29.0 | 27.5                  | 48.7 | 24.2 | 48.4 | 5.8                   | 50   | 0.488 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 13.7                   | 84.8            | 68.6 | 80.9 | 66.6 | 77.5 | 65.5 | 70.4                | 81.3 | 68.4 | 77.6 | 66.8 | 74.9 | 6.0                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.6                            | 98.0 | 74.8 | 65.0 | 92.7 | 73.0 | 63.6 | 87.9 | 71.1 | 34.3          | 81.5 | 32.7 | 77.4 | 31.3 | 74.8 | 82.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |     |            |     |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-----|------------|-----|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |     | n=50 years |     |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max | Min        | Max |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n) | (o)        | (p) |      |     |
| (4) 20.4          | 17.4 | 15.2 | DB                         | 19.4 | 91.8               | 5.8 | 2.6                                                | 15.3 | 93.6       | 11.9 | 95.1       | 8.7 | 96.6       | 4.5 | 98.5 | (4) |
| (5)               |      |      | WB                         | 18.6 | 74.2               | 5.6 | 2.0                                                | 14.6 | 75.6       | 11.4 | 76.7       | 8.2 | 77.8       | 4.2 | 79.3 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 53.0   | 39.9  | 41.2  | 45.7  | 51.5  | 57.7  | 62.4  | 66.4  | 66.0  | 61.0  | 54.3  | 47.1  | 41.6 | (6)  |
| (7)  |                                                                                  | DBStd       | 11.21  | 7.10  | 6.86  | 6.09  | 6.46  | 6.52  | 5.55  | 5.34  | 4.76  | 4.78  | 5.88  | 6.36  | 7.14 | (7)  |
| (8)  |                                                                                  | HDD50       | 1212   | 316   | 253   | 159   | 56    | 8     | 0     | 0     | 0     | 0     | 25    | 125   | 270  | (8)  |
| (9)  |                                                                                  | HDD65       | 4633   | 778   | 668   | 599   | 407   | 243   | 115   | 45    | 42    | 140   | 333   | 537   | 726  | (9)  |
| (10) |                                                                                  | CDD50       | 2294   | 3     | 5     | 24    | 101   | 247   | 372   | 509   | 496   | 330   | 160   | 38    | 9    | (10) |
| (11) |                                                                                  | CDD65       | 242    | 0     | 0     | 0     | 2     | 18    | 37    | 90    | 73    | 19    | 3     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH74       | 1605   | 0     | 0     | 0     | 17    | 152   | 287   | 579   | 466   | 97    | 7     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH80       | 409    | 0     | 0     | 0     | 1     | 25    | 77    | 158   | 132   | 16    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 7.3    | 8.6   | 8.4   | 8.0   | 7.3   | 6.8   | 6.6   | 6.5   | 6.2   | 6.2   | 6.6   | 7.8   | 8.3  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 32.1   | 2.7   | 2.4   | 1.9   | 1.7   | 2.3   | 2.7   | 3.1   | 3.5   | 2.7   | 2.8   | 3.2   | 3.2  | (15) |
| (16) |                                                                                  | PrecMax     | 38.1   | 5.5   | 4.9   | 5.0   | 3.5   | 4.1   | 5.0   | 6.3   | 10.3  | 8.1   | 5.5   | 5.7   | 7.1  | (16) |
| (17) |                                                                                  | PrecMin     | 24.1   | 0.2   | 0.6   | 0.1   | 0.0   | 0.8   | 0.8   | 0.7   | 1.0   | 0.5   | 0.9   | 0.2   | 0.7  | (17) |
| (18) |                                                                                  | PrecStd     | 3.7    | 1.3   | 1.1   | 1.1   | 1.0   | 0.9   | 1.2   | 1.5   | 2.2   | 1.7   | 1.0   | 1.3   | 1.7  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 55.8  | 57.2  | 66.2  | 76.4  | 83.1  | 87.7  | 90.2  | 88.8  | 81.9  | 74.1  | 63.8  | 57.6 | (19) |
| (20) |                                                                                  |             | MCWB   | 52.2  | 50.0  | 54.4  | 59.4  | 66.1  | 69.2  | 69.8  | 71.6  | 67.3  | 64.5  | 58.2  | 53.9 | (20) |
| (21) |                                                                                  | 2%          | DB     | 53.6  | 54.4  | 61.0  | 71.0  | 78.2  | 81.2  | 84.9  | 83.7  | 76.1  | 68.5  | 60.1  | 55.4 | (21) |
| (22) |                                                                                  |             | MCWB   | 50.7  | 49.4  | 52.1  | 57.4  | 62.8  | 66.2  | 68.3  | 69.2  | 65.2  | 61.7  | 56.2  | 52.4 | (22) |
| (23) |                                                                                  | 5%          | DB     | 51.4  | 52.1  | 57.3  | 66.0  | 73.4  | 76.6  | 80.7  | 79.1  | 72.0  | 64.8  | 57.8  | 53.4 | (23) |
| (24) |                                                                                  |             | MCWB   | 48.6  | 48.4  | 50.4  | 54.9  | 60.9  | 64.8  | 66.9  | 67.1  | 63.4  | 59.7  | 54.3  | 50.7 | (24) |
| (25) |                                                                                  | 10%         | DB     | 49.5  | 50.4  | 54.5  | 61.6  | 68.8  | 72.5  | 76.8  | 75.2  | 68.8  | 62.4  | 55.5  | 51.0 | (25) |
| (26) |                                                                                  |             | MCWB   | 46.8  | 47.3  | 49.3  | 52.8  | 58.9  | 62.5  | 65.9  | 65.1  | 61.6  | 58.3  | 52.7  | 48.5 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 53.0  | 52.1  | 55.9  | 61.6  | 68.9  | 71.2  | 72.7  | 73.6  | 69.5  | 65.2  | 59.8  | 54.9 | (27) |
| (28) |                                                                                  |             | MCD B  | 54.7  | 55.4  | 63.1  | 73.8  | 79.4  | 84.4  | 86.0  | 85.4  | 79.6  | 72.4  | 62.4  | 56.7 | (28) |
| (29) |                                                                                  | 2%          | WB     | 51.1  | 50.4  | 53.3  | 58.3  | 64.8  | 67.8  | 70.3  | 70.3  | 66.4  | 62.5  | 56.7  | 53.0 | (29) |
| (30) |                                                                                  |             | MCD B  | 53.2  | 53.0  | 58.7  | 67.9  | 74.2  | 77.8  | 80.9  | 80.8  | 73.4  | 67.2  | 59.5  | 55.1 | (30) |
| (31) |                                                                                  | 5%          | WB     | 49.0  | 49.1  | 51.5  | 56.0  | 62.3  | 65.5  | 68.3  | 68.1  | 64.5  | 60.5  | 54.8  | 50.9 | (31) |
| (32) |                                                                                  |             | MCD B  | 51.2  | 51.6  | 55.9  | 63.9  | 70.6  | 74.2  | 77.6  | 76.3  | 70.4  | 64.2  | 57.4  | 53.2 | (32) |
| (33) |                                                                                  | 10%         | WB     | 47.0  | 47.4  | 49.9  | 53.9  | 59.9  | 63.6  | 66.6  | 66.3  | 62.8  | 58.8  | 52.8  | 48.7 | (33) |
| (34) |                                                                                  |             | MCD B  | 49.2  | 50.2  | 53.9  | 60.3  | 67.3  | 71.1  | 74.9  | 73.4  | 67.7  | 61.9  | 55.2  | 50.8 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.2   | 8.3   | 11.0  | 13.9  | 14.1  | 14.1  | 13.7  | 14.2  | 12.4  | 10.7  | 7.7   | 6.7  | (35) |
| (36) |                                                                                  |             | MCDBR  | 8.2   | 9.5   | 15.9  | 21.0  | 22.1  | 21.9  | 20.9  | 21.7  | 17.7  | 13.9  | 9.2   | 7.5  | (36) |
| (37) |                                                                                  |             | MCWBR  | 7.4   | 7.3   | 9.5   | 9.7   | 9.8   | 10.3  | 8.7   | 9.9   | 8.5   | 8.3   | 7.1   | 6.7  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 8.0   | 8.6   | 13.4  | 18.2  | 18.7  | 18.8  | 18.1  | 19.3  | 15.4  | 12.5  | 8.8   | 7.6  | (38) |
| (39) |                                                                                  |             | MCWBR  | 7.6   | 7.4   | 9.1   | 9.4   | 9.8   | 10.0  | 8.8   | 10.0  | 8.2   | 8.1   | 7.2   | 7.2  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.334  | 0.354 | 0.390 | 0.417 | 0.426 | 0.423 | 0.426 | 0.417 | 0.393 | 0.370 | 0.346 | 0.330 | (40) |      |
| (41) |                                                                                  | taud        | 2.370  | 2.342 | 2.243 | 2.201 | 2.220 | 2.250 | 2.247 | 2.292 | 2.348 | 2.398 | 2.378 | 2.340 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 214    | 242   | 253   | 260   | 263   | 265   | 261   | 258   | 250   | 233   | 206   | 193   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 20     | 27    | 35    | 41    | 42    | 42    | 41    | 37    | 32    | 25    | 20    | 18    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 244    | 463   | 846   | 1371  | 1614  | 1750  | 1653  | 1410  | 1045  | 604   | 285   | 187   | (44) |      |
| (45) |                                                                                  | RadStd      | 24     | 87    | 111   | 134   | 128   | 142   | 176   | 123   | 79    | 51    | 30    | 24    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   | (46) |
|      |                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |      |
| (47) | Regional (77 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | (47) |

Nomenclature: See separate page